Telecommunications networks

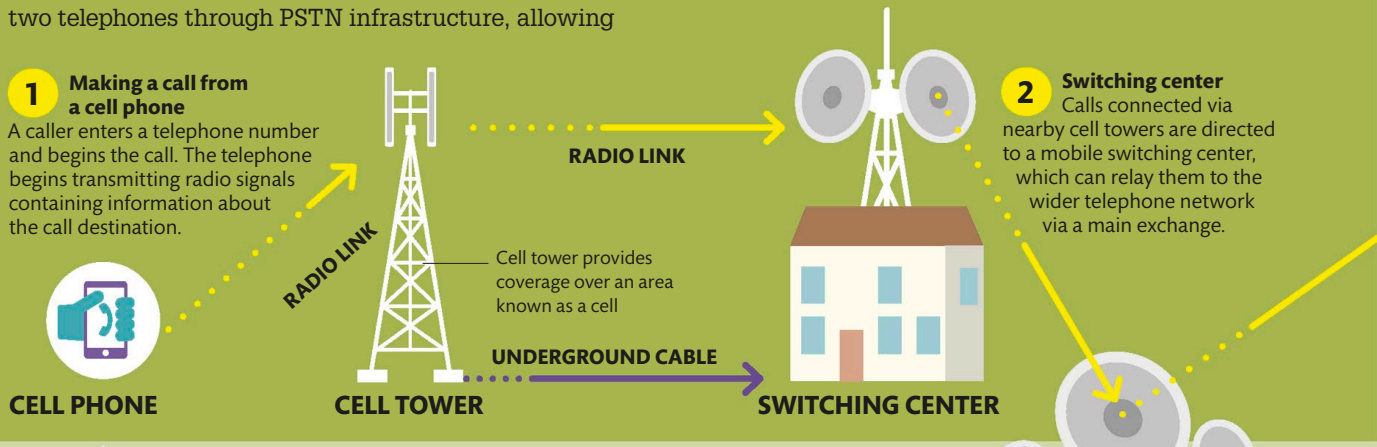
Telecommunications networks are systems that enable the exchange of information, including Internet traffic, across great distances. These networks are made up of connected points that relay signals through a system of wires, cables, satellites, and other infrastructure to reach their destination.

WHAT WAS THE FIRST TELECOMMUNICATIONS NETWORK?

The telegraph network was the first to enable long-distance communication. The first transatlantic cable was completed in 1858.

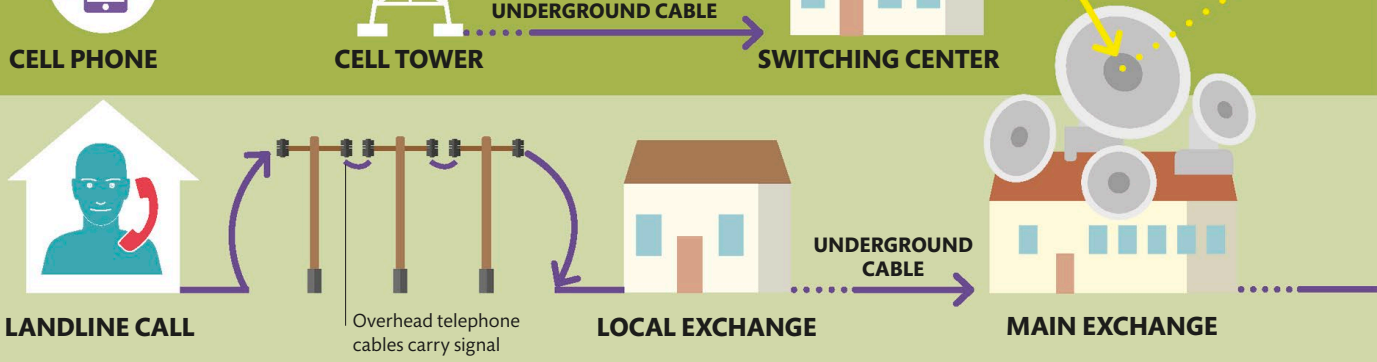
The telephone network

In the early days of the telephone, phones had to be permanently wired together for the callers to speak to each other. Now, they are connected to the public switched telephone network (PSTN) instead. During a call, a temporary connection is established between the two telephones through PSTN infrastructure, allowing for the high-speed exchange of sound information. This huge network is made up of the world's local, national, and regional telephone networks, linked with exchanges to allow communication between most telephones.



1 Making a call from a cell phone
A caller enters a telephone number and begins the call. The telephone begins transmitting radio signals containing information about the call destination.

2 Switching center
Calls connected via nearby cell towers are directed to a mobile switching center, which can relay them to the wider telephone network via a main exchange.



1 Outgoing landline call
A caller picks up the handset, making an electrical connection to the local exchange. As the caller enters a telephone number, signals indicating the call destination are sent down the line.

2 Local exchange
A local exchange connects telephones in a local area. If it detects a call destination that is further away, it relays the call to a main exchange.

3 Main exchange
Nonlocal cell phone and landline calls are passed to a main telephone exchange, which is capable of directing calls across much greater distances.